Reg. No. | | | | | | | | | | |

[12CE71]

SEVENTH SEMESTER B.E EXAMINATION, DECEMBER -2015

STRUCTURAL ANALYSIS-II

Time: 3 Hours **Maximum Marks:70**

Instructions: 1. Answer any FIVE FULL Questions.

2. Missing data if any may be assumed suitably.

1. Analyse the continuous beam loaded as shown in figure 1. By Kani's method. **14**

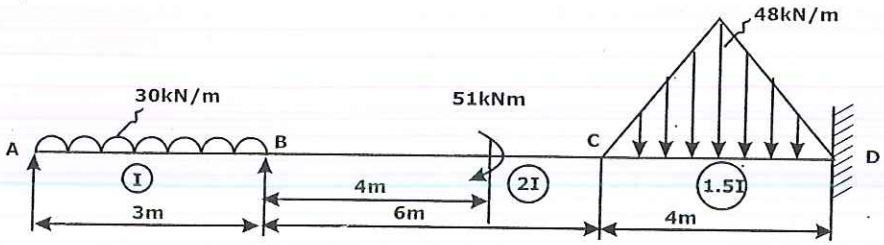

Fig. 1

2. Analyse the unsymmetrical frame shown in figure 2 by column analogy method. **14**

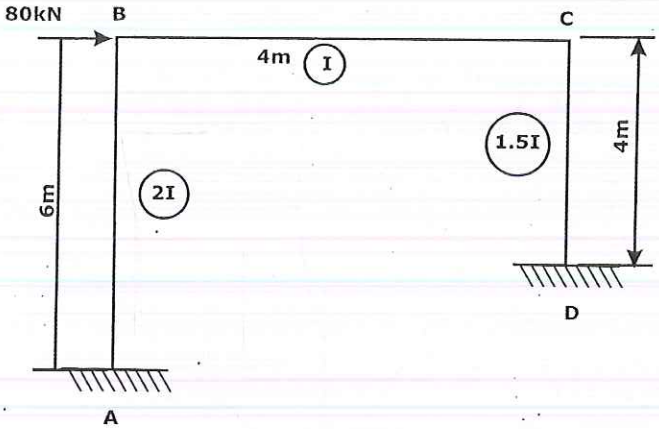

Fig. 2

3. **a.** Using portal method, analyse the building frame subjected to horizontal forces (due to wind load) as shown in figure 3. **07**

b. Analyse the portal frame given in figure. 3 by using cantilever method. **07**

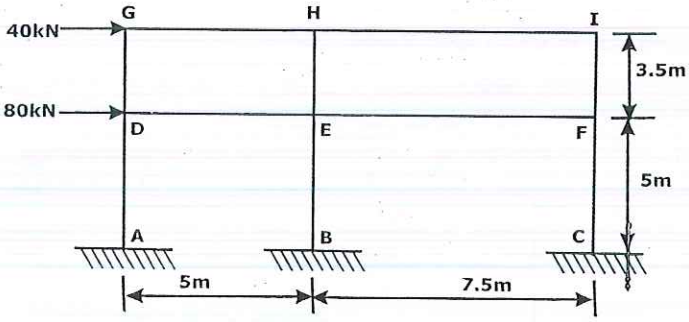

Fig. 3

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4. a. State and explain Maxwell's and Bette's reciprocal deflections theorem. **06**

b. Derive an expression for strain energy in element due to **08**

i. Axial force

ii. Bending moment

5. Analyse the portal frame as shown in figure 4 using flexibility matrix method. EI is constant. **14**

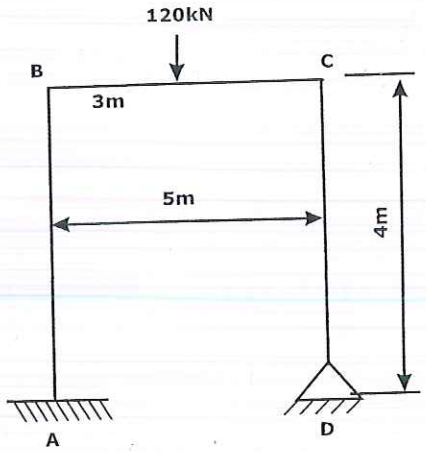

Fig. 4

6. Analyse the truss as shown in figure 5 by using stiffness matrix method. Take area of cross section for all members = 1000mm^2. **14**

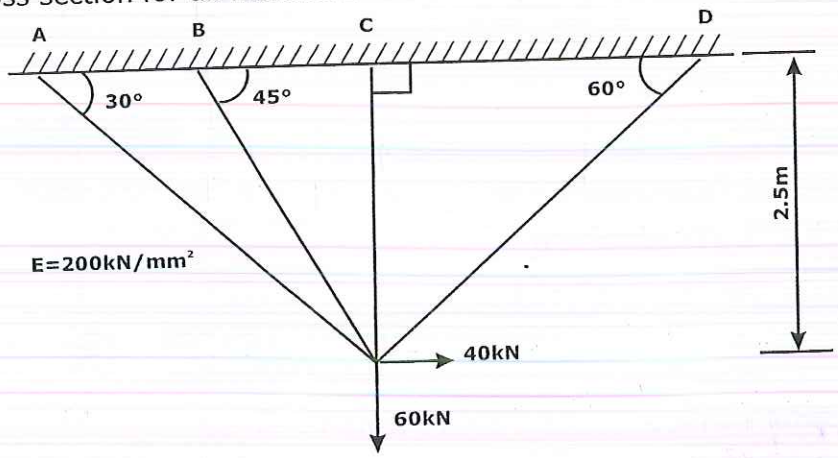

Fig. 5

7. Determine collapse load in the fixed beam as shown in figure 6 in which plastic moment capacity is $2M_P$ in one half and M_P in the other – half. **14**

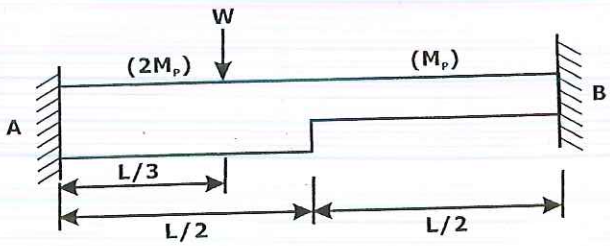

Fig. 6

8. **a.** Define the following **08**

i. Free vibration

ii. Forced vibration

iii. Damping

iv. Degree of freedom.

b. Consider the system as shown in figure 7, If K_1=2000N/W, K_2=1500N/W, K_3=3000 N/m. Find the mass, if the system has natural frequency of 50Hz. **06**

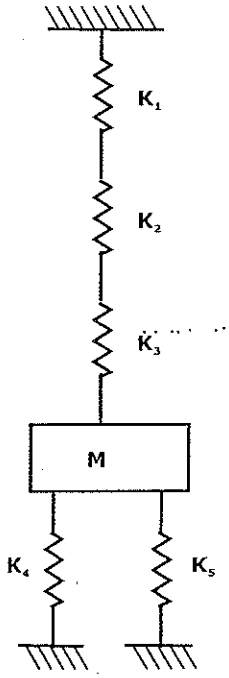

Fig. 7